

Enhancement Two Narrative

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For the algorithms and data structures enhancement, I selected the Corner Grocer application. I developed this C++ program for my final project in the SNHU Programming Languages course, which I took two years ago. In the original version of this application, the program would begin by reading a text file containing purchase history data for the produce section of Corner Grocer. The program would load the name of each unique item into a vector and load the name of each unique item and how many times it had been purchased into a map. By interacting with a menu, the user could accomplish two tasks: view how many times a particular produce item was purchased or view a complete list of how many times each produce item was purchased. When viewing the complete list, the user could choose whether the number of times each item was purchased would be represented by digits or a histogram.

I chose to include this artifact in my portfolio because I can demonstrate growth in my skills and abilities in algorithms and data structures. After all three enhancements are complete, the final application will use a non-relational database instead of reading data from a text file. As a result of this modification, new data structures must be created to effectively and efficiently access and loop through the Corner Grocer's sales data, creating the perfect opportunity for me to showcase my skills and abilities in algorithms and data structures.

Within Corner Grocer's sales database, the produce purchase history data is stored within a collection made up of documents with a key-value structure. This data structure is an appropriate choice because this application will be using MongoDB, which has a non-relational structure that provides flexibility and scalability. Additionally, MongoDB stores data internally in a binary JSON (BSON) format, so using JSON to represent the data being input and retrieved

from the database allows the MongoDB driver to easily convert back and forth between JSON and BSON (MongoDB Inc., n.d.b).

To streamline database interaction, I created a database class to manage database connections and define database queries. This class contains a query that reads data from the database into a cursor and passes it back to the main code. If the query returns a single item, the application logic accesses the fields containing the item name and the number of items purchased and prints out these values. If the query returns all items in the collection, the application logic iteratively accesses cursor contents, accessing the fields containing the item name and the number of items purchased and prints these values. I chose a cursor as the appropriate data structure to optimize querying, improve time complexity and efficiency, and improve overall performance; a cursor is more efficient for larger, non-relational datasets and uses less memory, so it optimizes the resources available and avoids a single, long wait time (MongoDB Inc., n.d.a). This enhancement showcases my understanding of the data structures required for application logic to work with documents stored in a non-relational database in an optimized and efficient way. When put together with my enhancements in the software design and engineering and databases categories, I will demonstrate my ability to create a cohesive and fully functional software application designed for long-term use.

I did meet the majority of the course outcomes that I planned to meet for the algorithms and data structures enhancement. I implemented database queries that require no or very little user input to retrieve the data the user wants to see on the front-end. Queries that do require user input only require the user to input the name of the item they want to lookup; they do not have to type out a perfectly formatted query. This will allow users to leverage the application to accomplish their goals through user-friendly means, meeting the first course outcome. I also

demonstrated my ability to evaluate the optimization, time complexity, and efficiency of my chosen data structures and algorithms and the trade-offs that come with using them, meeting the third course outcome. I began implementing the fifth course outcome through writing secure data queries. I still have some work to do in my next enhancement to ensure the database meets security best practices. Finally, my in-line comments and this narrative demonstrate my ability to develop and deliver professional written communication, which shows progress in meeting course outcome two. I will continue meeting course outcomes two and five in my next enhancement.

I faced a few challenges while working on the algorithms and data structures enhancement. For example, after creating my database and populating it with test data, I had trouble getting authenticated so that I could use the mongo shell to query and view the provided test data. In the end, I had to update my password in order to properly authenticate as something went wrong when I initially configured the credentials. Another challenge I faced was writing my own query in Python to read data. It took me a few tries to get the formatting correct so that I was passing the correct information into the find query.

While working on this enhancement, I solidified my understanding of creating and querying a non-relational database and working with cursors. I found the MongoDB documentation on PyMongo extremely helpful for setting up my database and refreshing my memory on how to properly connect to my database in a Python file. I conducted some research, including referencing assignments from previous classes, to refresh myself on how to set up a database class to streamline database interaction. I also learned how to be more concise in how I write code. During the development process for the enhancement, I reviewed my code multiple

times and made many improvements to condense and optimize the lines of code used to interact with the database and iterate through cursor elements.

References

MongoDB, Inc. (n.d.a). Access Data From a Cursor. MongoDB Docs.

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